

Outline

1. Motivation for SAVI

- 1.1 Problem with peeking at p-values
- 1.2 Wald's Sequential Probability Ratio Test

2. Validity: e-processes under $\mathcal{P}$

- 2.1 Setup & definitions
- 2.2 Martingales, test (super)martingales & e-processes
- 2.3 Optional stopping & Ville's inequality

3. Efficiency: e-processes under $\mathcal{Q}$

- 3.1 Simple $\mathbb{P}$ vs. simple $\mathcal{Q}$
- 3.2 Simple $\mathbb{P}$ vs. composite $\mathcal{Q}$
- 3.3 Composite $\mathcal{P}$ vs. composite $\mathcal{Q}$: Testing by betting

4. Further discussions

5. Summary

From validity to efficiency

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$\implies \mathbb{E}^{\mathbb{P}}[E_t] = 0.5(1 + \kappa) + 0.5(1 - \kappa) = 1$, so E_t is an e-variable for $\mathbb{P}$

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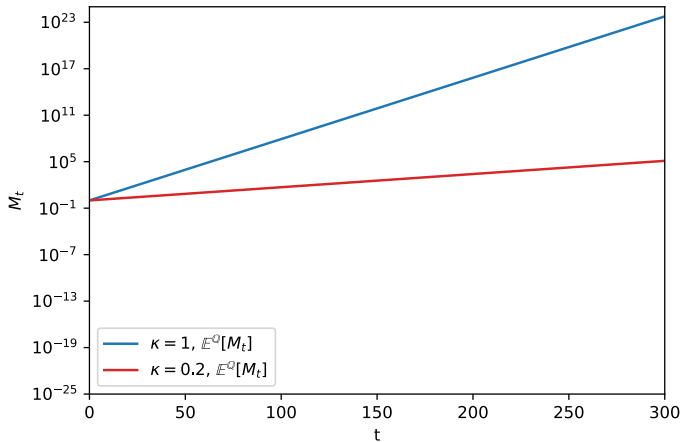
- $\mathbb{E}^{\mathbb{Q}}[M_t] = t\mathbb{E}^{\mathbb{Q}}[E_1] = t(1 + 0.2\kappa)$ is maximised at $\kappa = 1$

Maximise $\mathbb{E}^{\mathbb{Q}}[M_t]$ for efficiency?

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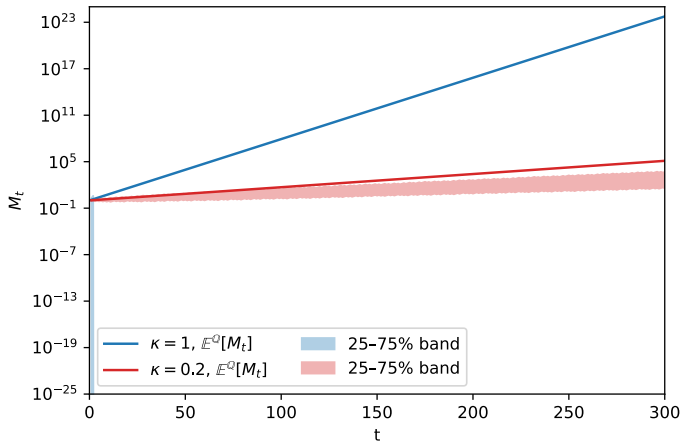


Mean path: In $\kappa = 1$ case, only the all-ones path $X_1, \dots, X_t = 1$ contributes to $\mathbb{E}^{\mathbb{Q}}[M_t]$ with tiny probability 0.6^t

Figure: $M_t = \prod_{i=1}^t E_i$ under $\mathbb{Q}$, with 6000 runs.

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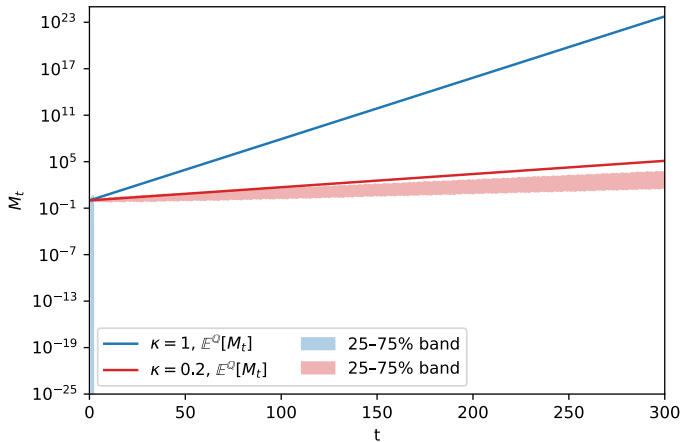


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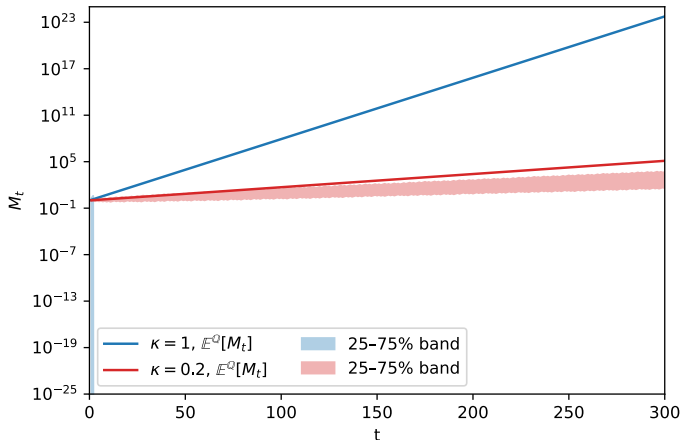
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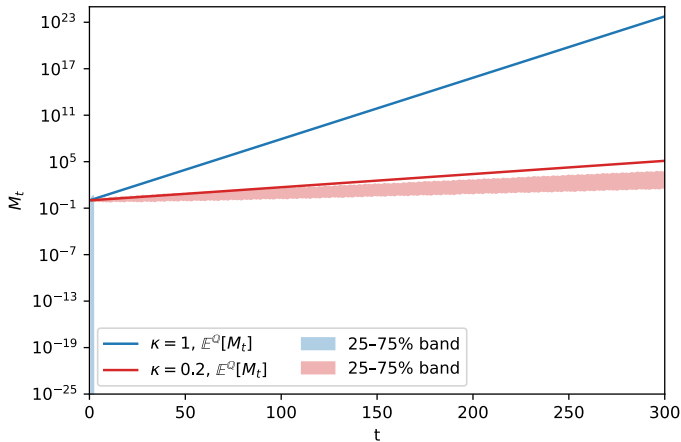
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$$\frac{1}{t} \log M_t = \frac{1}{t} \log \prod_{i=1}^t E_i$$

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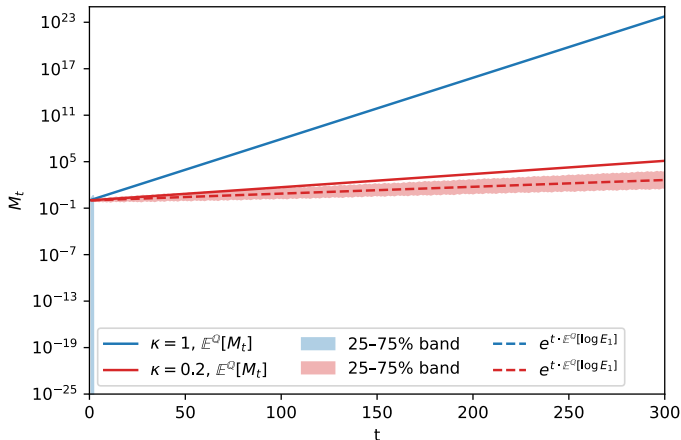


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$$\Rightarrow \text{median}(M_t) \sim e^{t \mathbb{E}^{\mathbb{Q}}[\log E_1]}$$

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- Evidence multiplies (~~sums~~)
 - $\Rightarrow$ **Typical path** governed by geometric (~~arithmetic~~) mean
 - $\Rightarrow$ Look at logs
- $\mathbb{E}^Q[\log M_t]$: e-power/expected log-growth

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Simple $\mathbb{P}$ vs. simple $\mathbb{Q}$

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The Likelihood Ratio (LR) process M^* given by $M_0^* = 1$ and $M_t^* = \frac{d\mathbb{Q}|_{\mathcal{F}_t}}{d\mathbb{P}|_{\mathcal{F}_t}}(X_1, \dots, X_t)$ for $t \geq 1$ is a test martingale for $\mathbb{P}$.

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Theorem (Log-optimality)

For any stopping time τ that is **finite** $\mathbb{Q}$ -a.s. and any e-process M for $\mathbb{P}$:

$$\mathbb{E}^{\mathbb{Q}}[\log M_{\tau}^*] \geq \mathbb{E}^{\mathbb{Q}}[\log M_{\tau}].$$

Proof. For fixed t , setting $M_t = M_t^*$ maximises $\mathbb{E}^{\mathbb{Q}}[\log M_t]$ (presented by François) + Reduction

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Next: Going back to our Bernoulli example...

Simple $\mathbb{P}$ vs. simple $\mathbb{Q}$: log-optimality of LR process

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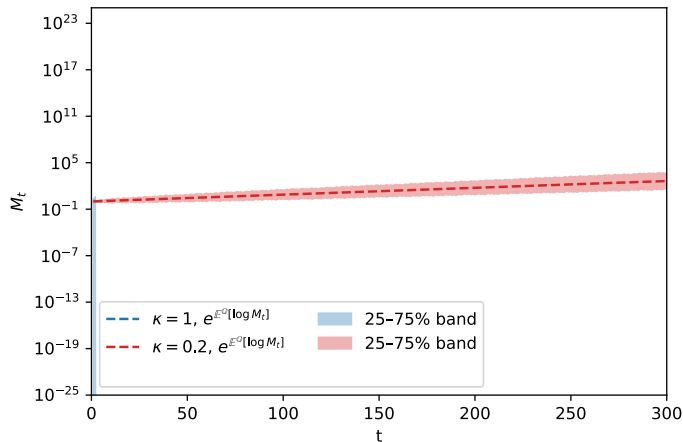
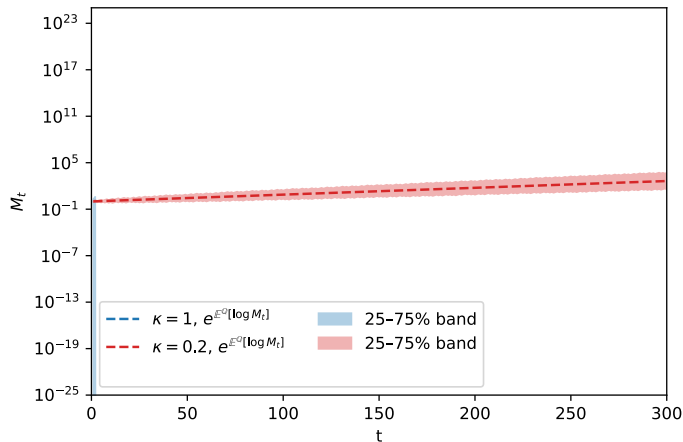


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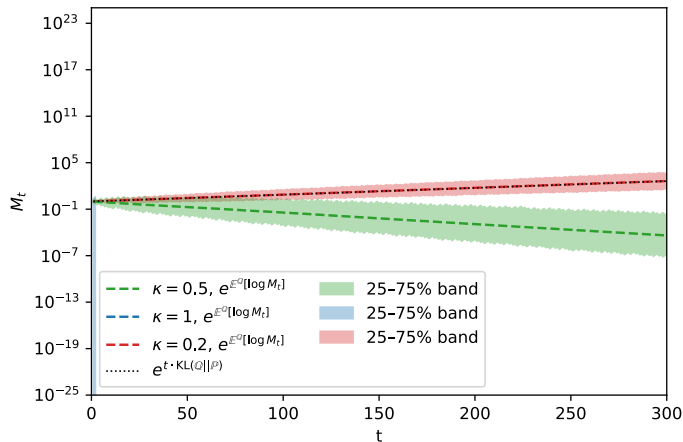


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Remarks:

- For every fixed t ,

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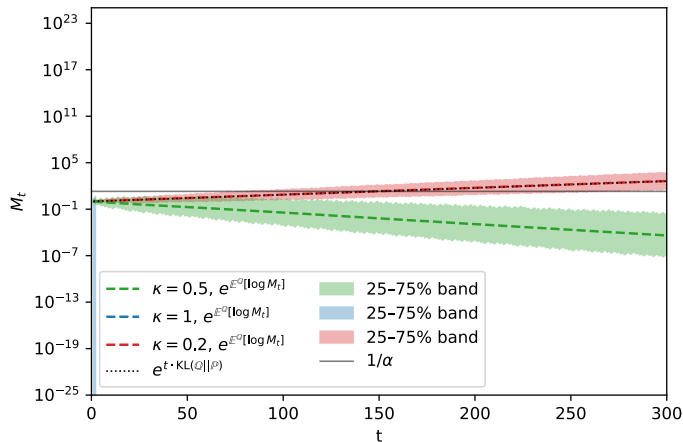


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- Asymptotically,

$$\text{Power } \mathbb{Q}(M_t \geq \frac{1}{\alpha}) \xrightarrow[t \rightarrow \infty]{} 1$$

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Definition (Asymptotic log-optimality)

An e-process M is asymptotically log-optimal for $\mathbb{P}$ against $\mathcal{Q}$ if for every $\mathbb{Q} \in \mathcal{Q}$,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \left(\log M_t - \log M_t^{\mathbb{Q}} \right) \geq 0 \quad \text{in } L^1\text{-convergence under } \mathbb{Q}$$

where $M^{\mathbb{Q}}$ is the oracle LR process of $\mathbb{Q}$ to $\mathbb{P}$.

- Requires at least the same long-run log-growth rate as $M^{\mathbb{Q}}$
- Covers any e-process M that grows an $e^{o(t)}$ factor slower than $M^{\mathbb{Q}}$

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Definition (Consistency)

An e-process M is said to be consistent against $\mathcal{Q}$ if $M_t \rightarrow \infty$, $\mathbb{Q}$ -a.s. as $t \rightarrow \infty$.

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Example: Testing $\mathbb{P}$ against $\mathcal{Q} = \{\mathbb{Q}_\theta : \theta \in \Theta_1\}$ with iid data.

Plug-in LR: Set $M_0 = 1$, and for $t \geq 1$ use $M_t = \prod_{i=1}^t \frac{q_{\hat{\theta}_{i-1}}(X_i)}{p(X_i)}$ with predictable $\hat{\theta}_{i-1} \in \Theta_1$

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Proposition (informal)

Plug-in is asymptotically log-optimal when $\theta_i \rightarrow \theta$ under $\mathbb{Q}_\theta$ in a suitable sense, given log-LR is concave, score function has bounded variance.

Example: Given iid data from $N(\theta^\dagger, 1)$, goal is to test $\mathcal{H}_0 : \theta^\dagger = 0$ vs $\mathcal{H}_1 : \theta^\dagger > 0$. For illustration, take $\theta^\dagger = 0.3$.

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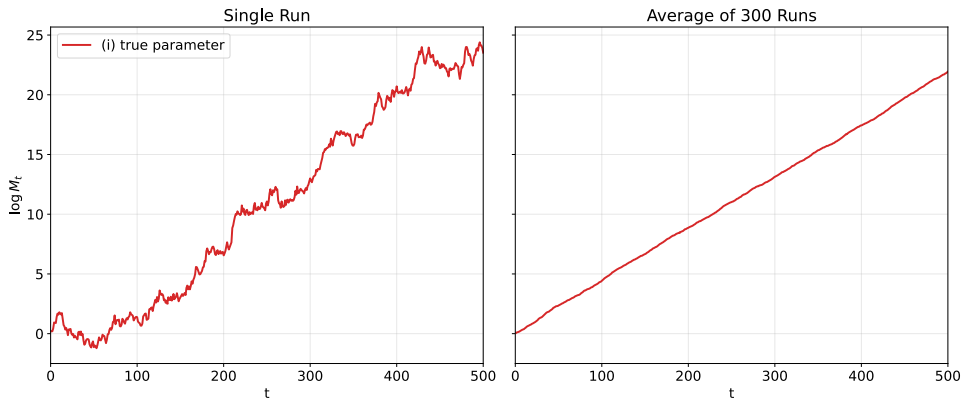


Figure: Several ways of constructing e-processes from LR processes.

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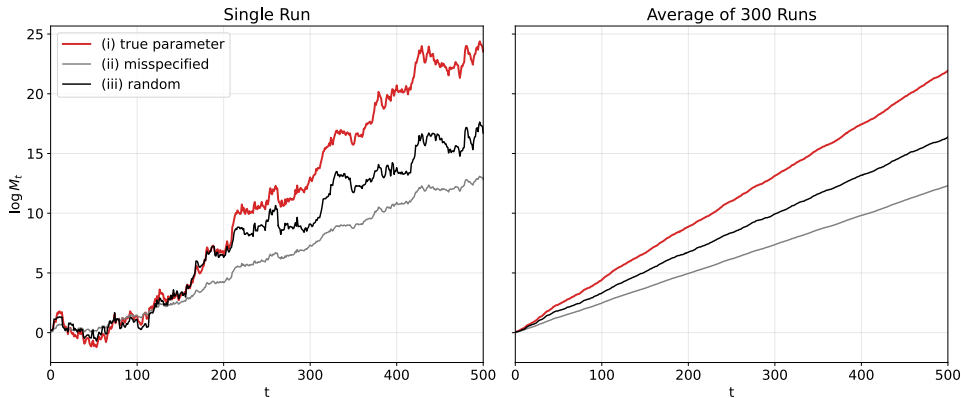


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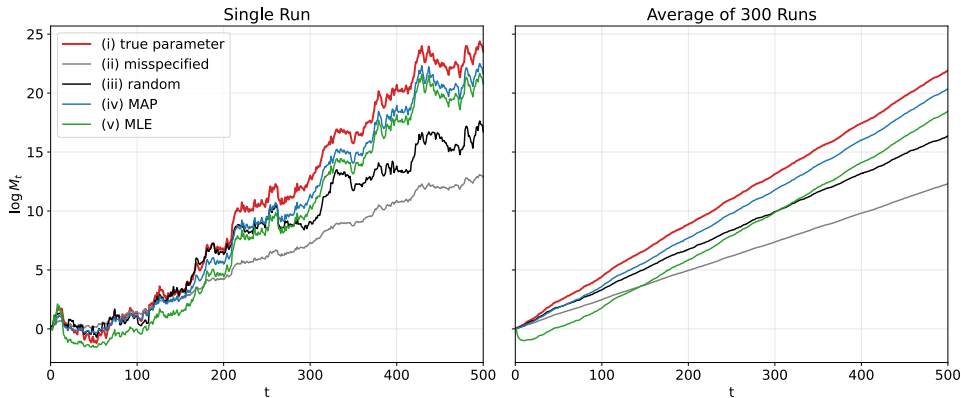


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- (iv) MAP: choose θ_i by the MAP estimator with prior $\theta \sim N(0.1, 0.2^2)$
- (v) MLE: choose θ_i with $\theta_1 := 0.1$ and θ_i the MLE of θ based on $X_1, \dots, X_{i-1}$

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Initialize wealth $M_0 = 1$.

For $t = 1, 2, \dots$:

- Declare a bet $E_t : \mathcal{X} \rightarrow [0, \infty)$ with $\mathbb{E}^{\mathbb{P}}[\textcolor{red}{E}_t(X_t) \mid \mathcal{F}_{t-1}] \leq 1 \quad \forall \mathbb{P} \in \mathcal{P}$.
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Proposition

If $\mathbb{E}^{\mathbb{P}}[E_t \mid \mathcal{F}_{t-1}] \leq 1$ for all $\mathbb{P} \in \mathcal{P}$ and $t \geq 1$, then $M_t = \prod_{s=1}^t E_s$ for $t \geq 1$ with $M_0 = 1$ is a test supermartingale (hence e-process) for $\mathcal{P}$.

Proof. $\mathbb{E}^{\mathbb{P}}[M_t \mid \mathcal{F}_{t-1}] = M_{t-1} \mathbb{E}^{\mathbb{P}}[E_t \mid \mathcal{F}_{t-1}] \leq M_{t-1}$ for every $\mathbb{P} \in \mathcal{P}$.

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Initialize wealth $M_0 = 1$.

For $t = 1, 2, \dots$:

- Declare a bet $E_t : \mathcal{X} \rightarrow [0, \infty)$ with $\mathbb{E}^{\mathbb{P}}[\textcolor{red}{E}_t(X_t) \mid \mathcal{F}_{t-1}] \leq 1 \quad \forall \mathbb{P} \in \mathcal{P}$.
 - Observe data X_t .
 - Update wealth: $M_t = M_{t-1} \cdot \textcolor{red}{E}_t(X_t) = \prod_{s=1}^t E_s(X_s)$.
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 - (ii) **Compromise:** Avoid all-in; pick stake $\lambda_t \in [0, 1]$ to hedge misspecification.

Testing by betting (composite $\mathcal{P}$ vs. $\mathcal{Q}$)

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$(M_t)_{t \geq 0}$ is a test supermartingale (hence e-process) for $\mathcal{P}$.

Proof. $\mathbb{E}^{\mathbb{P}}[(1 - \lambda_t) + \lambda_t E_t \mid \mathcal{F}_{t-1}] \leq (1 - \lambda_t) + \lambda_t \cdot 1 = 1$ for every $\mathbb{P} \in \mathcal{P}$.

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Next: How to optimise the stakes $(\lambda_t)_{t \geq 1}$?

Optimising predictable stakes $(\lambda_t)_{t \geq 1}$

Definitions

(i) For an alternative measure $\mathbb{Q}$, the **oracle** e-process built on $(E_t)_{t \geq 1}$ is $(M_t)_{t \geq 0}$ with

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Next: Going back to our iid Bernoulli example...

Example:

- $(X_t)_{t \geq 1}$ iid from $\text{Bern}(p)$, with $\mathcal{H}_0 : p \leq 0.5$ vs. $\mathcal{H}_1 : p \geq 0.55$.

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$$\implies \mathbb{E}^{\mathbb{P}}[E_t(X_t) \mid \mathcal{F}_{t-1}] \leq 0.5 \cdot 2 + 0.5 \cdot 0 = 1 \text{ for } p \leq 0.5.$$

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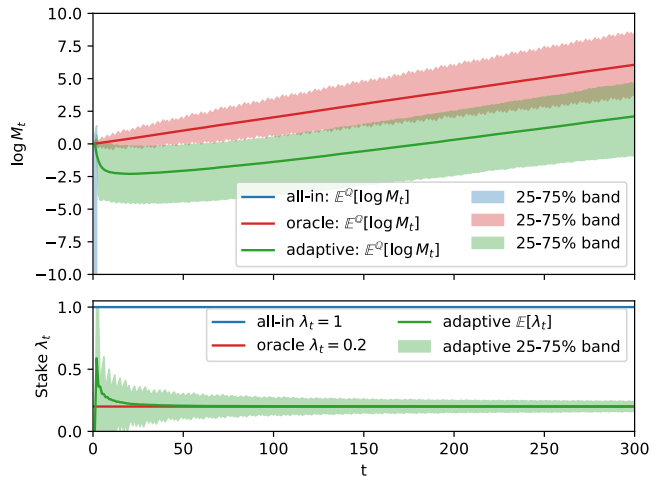
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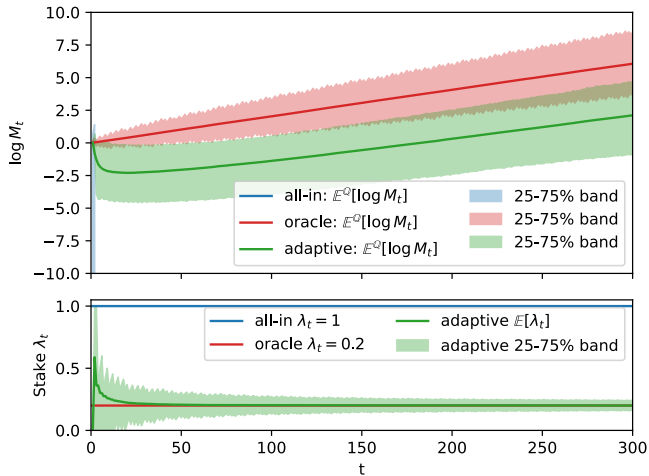
- Nature picks $\mathbb{Q} = \text{Bern}(0.6)$.
- **Oracle** e-process built on $(E_t)_{t \geq 1}$ bets with

$$\lambda_t = 0.2 \in \arg \max_{\lambda \in [0,1]} \mathbb{E}^{\mathbb{Q}}[\log((1 - \lambda) + \lambda E_t)].$$

- **Empirically adaptive** e-process bets with

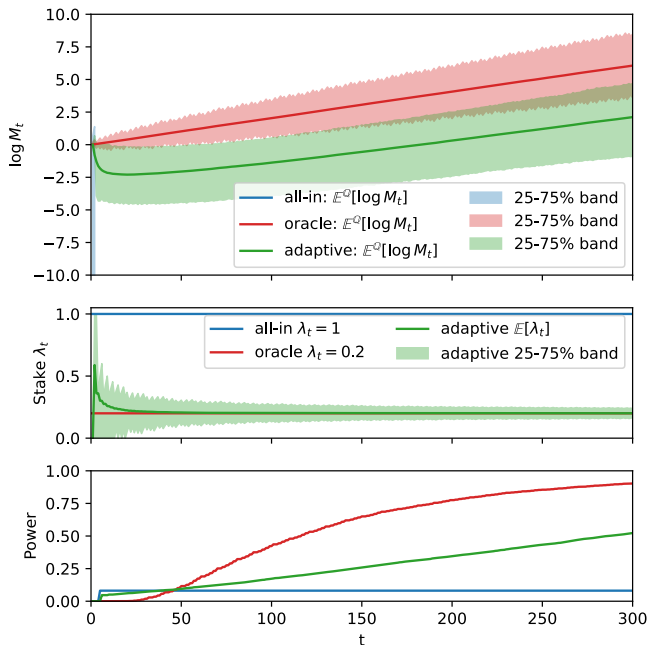
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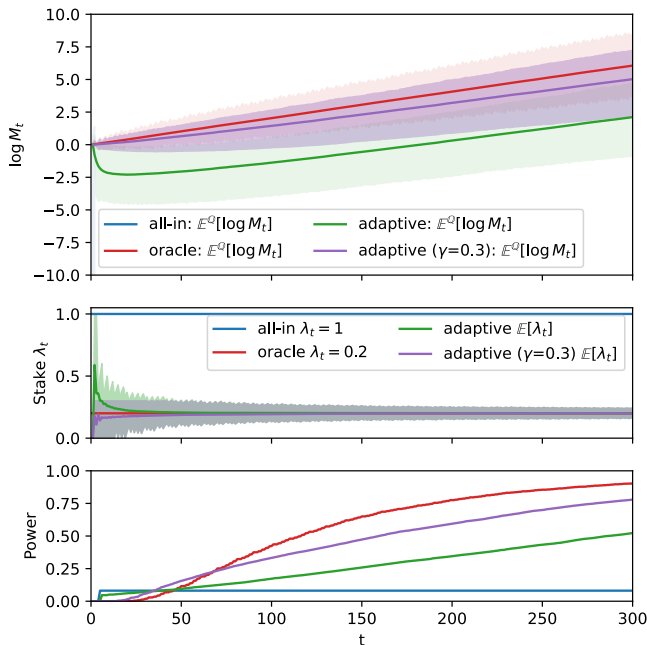
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 - has good e-power & power when $(E_t)_{t \geq 1}$ iid under $\mathbb{Q}$



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Empirically adaptive e-processes

Theorem

Let $(E_t)_{t \geq 1}$ be iid under the alternative distribution $\mathbb{Q}$ such that $\mathbb{E}^{\mathbb{Q}}[\log E_1]$ is finite. The empirically adaptive e-process $(M_t)_{t \geq 0}$ with $\gamma = 1$ satisfies the following:

(i) Asymptotic log-optimality in the sense that

$$\lim_{t \rightarrow \infty} \frac{1}{t} \left(\log M_t - \log M_t^{\mathbb{Q}} \right) \geq 0 \quad \text{in } L^1\text{-convergence under } \mathbb{Q}$$

with the oracle e-process $(M_t^{\mathbb{Q}})_{t \geq 0}$ built on $(E_t)_{t \geq 1}$.

(ii) Consistency, i.e., if $\mathbb{E}^{\mathbb{Q}}[E_1] > 1$, then $M_t \rightarrow \infty$ $\mathbb{Q}$ -a.s. as $t \rightarrow \infty$.

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Proof.

- (i) follows from LLN.

- (ii) due to for $E \geq 0$, $\mathbb{E}^{\mathbb{Q}}[E] > 1 \iff \exists \lambda \in [0, 1]$ s.t. $\mathbb{E}^{\mathbb{Q}}[\log((1 - \lambda) + \lambda E)] > 0$.